

Input, Process, Output

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

AI or Not?

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Finish the visitor example for Representations and information paths. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Representations and information paths. Proposed explanation: “Every output is the final action”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Investigate a new loss of information

A classroom device receives four scores and stores only their total, 20. A student claims that every score must be 5. Give two different lists of four nonnegative whole-number scores with total 20. Explain precisely what the stored total establishes and what it does not.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A library kiosk receives 'mystery', filters a catalogue by genre and displays three titles. No speaker is attached. Draw its information path. A visitor says it read three books aloud. Which stage contradicts that claim?

6. Solve the given case

A playlist helper receives a mood, matches song tags and displays three titles.

Name input, process and output. Is displaying titles the same as playing audio?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A classroom fan receives a temperature reading, compares it with 25 degrees C, and turns on. Name the input, process and output. What should happen if the reading is missing?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 2 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe

response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Draw the Classroom Sorter's first input-process-output map and show where a person may review the result.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

11. Transfer practice from the original programme

A library kiosk receives 'mystery', filters a catalogue by genre and displays three titles. No speaker is attached. Draw its information path. A visitor says it read three books aloud. Which stage contradicts that claim?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Trace a request

The kiosk receives genre=science and maximum pages=120. The catalogue contains A science 80 pages, B science 150 pages and C mystery 90 pages. It keeps records with the requested genre and no more than 120 pages. Draw the input, process and output, naming the surviving record.

My reasoning and evidence

13. W2 Finish the visitor example

A record stores only total visitors=3. Give two different arrival schedules that produce this total. Can the

total tell us the busiest hour? Explain.

My reasoning and evidence

14. W3 Choose a sufficient representation

For each task, choose the information needed: decide whether a book is under 120 pages; quote its first sentence exactly; decide whether a copy is available now. Explain why a genre tag alone is insufficient for all three.

My reasoning and evidence

15. W4 Investigate a new loss of information

A classroom device receives four scores and stores only their total, 20. A student claims that every score must be 5. Give two different lists of four nonnegative whole-number scores with total 20. Explain precisely what the stored total establishes and what it does not.

My reasoning and evidence

16. W5 Diagnose before repairing

A voice calculator receives a spoken request for 16 divided by 4. Its transcript says 60 divided by 4 and its calculator returns 15. Identify the first incorrect stage, show the intended result and propose a check before calculation.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record:

Evidence: First step I repaired and why

My record:

Evidence: New case checked later; date and result

My record:

Evidence: What I can now explain without notes

My record:

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Input, Process, Output

A fictional classroom cupboard records coloured pencils, then reports just a total.

Box A has 2 red and 4 blue pencils. Box B has 5 red and 1 blue.

The process adds colour counts and stores only the total, not the two counts.

AI ACADEMY / AI-1/2

Different colour counts, same total

	Red	Blue	Total
Box A	2	4	6
Box B	5	1	6

Read a cell using both its row and column labels.

Input, Process, Output

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read colours

A: red 2, blue 4. B: red 5, blue 1.



2. Add counts

A: $2+4=6$. B: $5+1=6$.



3. Store total

Both records become total 6.



4. Ask a new question

The saved totals cannot tell which box has more red pencils.

CHANGE ONE THING

Store red count alongside total.

Explain the changed result using the matching lesson.

Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Choose sorter inputs that retain the clues needed for its decision; draw the transformation and identify discarded details.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Choose sorter inputs that retain the clues needed for its decision; draw the transformation and identify discarded details.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional classroom cupboard records coloured pencils, then reports just a total.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The saved totals cannot tell which box has more red pencils.

1. Add counts
2. Read colours
3. Ask a new question
4. Store total

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Ask a new question The saved totals cannot tell which box has more red pencils. Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct inputs lead to the same stored output. Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally.

Practice 2: explain the deciding evidence

Changed-case practice: Store red count alongside total. Which conclusion is supported by the supplied facts?

1. The records become (6,2) and (6,5).
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The records become (6,2) and (6,5). Blue count can now be recovered by total minus red, assuming these are the only colours.

Repair a missing building block

Review AI or Not? (ai-1/1). Goal: Distinguish an observed result from an explanation of how it was produced. Short practice: Point to the learned stage. Does the changed display prove the weather forecast changed?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
